4.2 Linear Separability

We now turn our attention to what extent closed semantic regions exist in the embedding space. For a fixed word w that frequently occurs in SemCor, we sample up to $n = 500$ embedding vectors, apply 5-fold cross-validation, and oversample any imbalanced-class datasamples. The input is normalized, and we apply dimensionality reduction using PCA to k components, ensuring that each of the semantic classes contains at least 20 samples in the dataset. We use the SemCor semantic class labels as the response variables for the classification task. We only include semantic classes for the given word, leaving us with few class-labels. Results are shown in Table 1.

k	% variance	accuracy (mean / std)
10	0.30	0.74 / 0.05
20	0.44	0.80 / 0.04
30	0.54	0.82 / 0.03
50	0.70	0.87 / 0.04
75	0.79	0.83 / 0.04
100	0.85	0.89 / 0.03

Table 1: Average mean and standard deviation of the accuracy of a linear classifier trained on the 2 most common semantic classes for the words `was`, `one`, `is`. The choice of words is limited to the datasize of SemCor to allow for a significant size of datasamples.

Accuracy rates of over 75% are achieved with $k = 20$. The % *variance* refers to the explainable variance when the largest k eigenvalues are kept, as calculated by $\sum_i^k \sigma_i$ where σ_i is the ith largest eigenvalue, hinting to how much information according to the largest k principal components are kept. Similar results are achieved for 2-class and multi-class classification tasks with other words (see Appendix A.2). We conclude that the individual semantic classes are – to a reasonable extent – linearly separable. As such, contextual word embeddings are not randomly distributed over the embedding space, and closed semantic regions do form.

4.3 Polysemy vs. Variance

Before we analyze the structure of individual semantic classes, we want to understand how polysemy relates to the mean standard deviation of all contextual word embeddings X sampled for the word w. This helps us to understand how we need to adapt different clustering models.

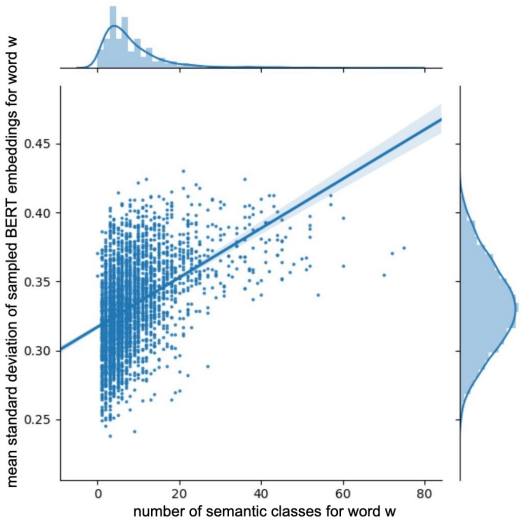

Figure 4: For each word w, we sample up to $n = 500$ contextual word embeddings X. We calculate the mean standard-deviation across embedding-dimensions as $\sum_i^n \sum_j^d {x_i}^j$ where $x_i^j \in \mathbf{R}^d$ is the jth dimension of the ith sampled embedding vector for word w. WordNet is used to retrieve the number of semantic classes of w, denoting the amount of polysemy of word w.

Figure 4 shows that polysemous words have high variance, an idea initially put forth by Miller and Charles (1991). As such, vectors of polysemous words seem to be distributed at least as dispersed around the space as non-polysemous words do. Notice that the converse is not true, as there are non-polysemous words that have high variance. Amongst others, these could include stopwords as hinted by Ethayarajh (2019).

4.4 Clusterability

We now want to understand to what extent distinct semantic clusters exist. For a set of words $w_1, \ldots, w_n$, we sample up to $n = 500$ embedding vectors per word from SemCor and the `news.2007.corpus` [2] and apply dimensionality reduction using PCA to k dimensions. Due to the limited size of SemCor, we set the words in the development set to `was`, `thought`, `made`, `only`, `central`, `pizza` and the set of words in the test set to `run`, `round`, `down`, `bank`, `key`, `arms`. We include both polysemous words, as well as words which have a single recorded WordNet meaning, such that our experiments do not overfit to polysemous words. With default-package hyperparameters, all clustering algorithms would indicate that no distinct clustering could be found,

[2] http://www.statmt.org/wmt14/training-monolingual-news-crawl/

i.e. the sampled word embeddings would form a continuous density. Because we want to see to what extent BERT conforms to commonly accepted linguistic senses as given by WordNet, we apply the NetworkX Chinese Whispers implementation (Hagberg et al., 2006) on the resulting graph. Hyperparameters and their respective bounds for all clustering models are listed in Appendix A.1. We include the `[SEP]` tag at the end of the sentence, as this increases performance on all clustering methods. The ARI is exclusively calculated on samples for which we have the ground-truth cluster label and stems from the mean of multiple such word clusterings. We notice that choosing suitable hyperparameters is non-trivial and thus apply automated model- and hyperparameter selection, making use of random search (Bergstra and Bengio, 2012) and bayesian optimization (Wang et al., 2013) [3].

Clustering Model	**ARI Score**
Affinity Propagation	0.316
Modified Chinese Whispers	**0.457**
DBScan	0.170
HDBScan	0.298
MeanShift	0.251

Table 2: The maximum ARI scores achieved during hyperparameter optimization on different models for $k = 20$ and $n = 1000$.

The models used and their maximal performance after 300 trials of hyperparameters search are recorded in Table 2. Our modified Chinese Whispers algorithm is the best-performing clustering model. However, with an ARI score of 0.457, this method is not able to perfectly distinguish between multiple WordNet semantic classes [4]. To understand why this is the case, we proceed with a qualitative evaluation of some resulting clusters. One such clustering is depicted in Table 3, presenting four partitions for `arms` [5]. We achieve similar such results for 9 other words but focus on one example for conciseness. Notice that the clusters differ not only in semantics but also in other linguistic phenomena, most notably sentiment.

Given the quantitative and qualitative evaluation, we conclude that one cannot generalize that a clear distinction between semantic concepts in contextual word embeddings produced by BERT exists. One of numerous counterexamples is underlined in the left visualization of Figure 1. Certain combinations of semantics, syntax, and sentiment are more frequent than others (Hagoort, 2003; May et al., 2019), likely affecting the subspace structure and sometimes resulting in clusters that are distinct due to their simultaneous difference in both semantic and syntactic features (see Appendix A.3). However, this work also poses the question to what extent rule-based and handcrafted notions of semantics, such as the ones given by WordNet, are appropriate, opening the question to what extent BERT actually encodes a more flexible notion of semantics that is not rooted in hard distinctions between senses. We leave analysis in this direction to future work.

Partition	Representative Sample
1	Ms. Gotbaum tried to slide her handcuffed **arms** from her back to her front ...
2	She swooped him up into her **arms** and kissed him madly ...
3	... and shuttle robotic **arms** of a solar array and truss ...
4	The classic years of the **arms** race, the 1950s and '60s before ...

Table 3: Representative samples for the clusters found by the best performing clustering model for the word `arms`. Partitions 1-3 consider a person's arms, whereas partition 4 considers `arms` as a synonym to `weaponry`. Partitions 1, 2 and 3 strongly contrast in sentiment (scared, loving, and confident respectively).

5 Conclusion

In this paper, we investigated how contextual word embeddings produced by BERT capture semantic concepts with a strong focus on polysemy. Our findings show that BERT creates closed semantic regions that are not clearly distinguishable from each other, seamlessly transitioning from one into another. We have shown that subspace organization is not purely determined by semantics. Instead, it is also intertwined with concepts such as syntax and sentiment. Finally, the repeated limitations of hard distinctions between senses as given via WordNet also open up the question to what extent BERT adds a more flexible notion of semantics, compared to the hard-coded examples formed by

[3] We use the implementation by https://github.com/facebook/Ax

[4] An ARI score of at least 0.7 is desirable to conclude a significant overlap between two clusters

[5] See Table 8 for a complete example clustering

linguists. A better understanding of these relations will be key to developing more interpretable and expressive word embeddings, as well as linguistic knowledge representations.

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